

Appendix G: Decoder Selected Configurations

A Reading the Tables

The tables report selected decoder-side configurations for Pythia-1.4B. Rows are transformer layers, indexed from 0 to 23. The layer-sparsity table reports which sub-blocks are kept in the verified 50% configuration evaluated with approximately 131k calibration tokens. The quantization and low-rank tables group the four repeated matrix families per layer: attention output, query-key-value projection, MLP down projection, and MLP up projection.

Layer-sparsity entries show the kept sub-blocks: attention plus MLP, attention only, MLP only, or full layer drop. **Quantization levels** are shown directly as bit-widths: 2, 2.5, 3, 4, or 8 bits. **Low-rank levels** are shown as the retained fraction of singular values: 5%, 10%, 20%, 35%, 50%, or 75%. Darker cells indicate larger retained precision or rank.

B Layer Sparsity: 50% Target, 131k Calibration Tokens

Layer	Kept	Layer	Kept
0	Attn.+MLP	12	Full drop
1	Attn. only	13	MLP only
2	Attn.+MLP	14	Attn. only
3	Attn.+MLP	15	MLP only
4	Attn.+MLP	16	MLP only
5	Attn.+MLP	17	Full drop
6	Attn.+MLP	18	Full drop
7	Attn.+MLP	19	Full drop
8	Full drop	20	MLP only
9	Full drop	21	Full drop
10	Attn.+MLP	22	Full drop
11	Attn.+MLP	23	Full drop

C Quantization: 2.25-Bit Target

L	Attn. out	QKV	MLP down	MLP up
0	3	3	3	3
1	4	4	3	3
2	3	3	3	4
3	4	3	3	3
4	3	3	3	4
5	4	3	3	4
6	3	3	3	4
7	3	4	3	4
8	4	4	3	4
9	4	4	3	4
10	4	3	3	3
11	3	4	3	3
12	3	4	3	3
13	4	4	3	3
14	3	4	3	3
15	3	8	3	3
16	4	3	3	3
17	3	4	3	3
18	3	3	3	3
19	4	3	3	3
20	4	3	3	3
21	3	3	3	3
22	4	3	3	3
23	3	3	3	3

D Quantization: 2.5-Bit Target

L	Attn. out	QKV	MLP down	MLP up
0	4	3	3	4
1	4	4	3	3
2	4	3	3	4
3	4	4	3	4
4	4	3	3	4
5	4	3	3	4
6	4	4	3	4
7	4	4	3	4
8	4	4	3	4
9	4	4	3	4
10	4	3	3	4
11	4	4	3	4
12	4	3	4	3
13	4	4	3	3
14	3	4	3	4
15	4	4	3	3
16	3	4	3	3
17	3	4	3	3
18	4	4	4	3
19	4	3	4	3
20	3	4	3	3
21	8	4	3	4
22	4	4	4	3
23	4	4	4	3

E Low-Rank: 0.5 Target

L	Attn. out	QKV	MLP down	MLP up
0	35%	20%	75%	75%
1	35%	35%	35%	50%
2	10%	5%	35%	35%
3	35%	35%	50%	50%
4	35%	35%	50%	50%
5	50%	50%	35%	50%
6	20%	35%	35%	50%
7	35%	50%	35%	50%
8	20%	35%	35%	50%
9	35%	35%	35%	50%
10	20%	35%	50%	75%
11	35%	75%	35%	50%
12	50%	50%	35%	35%
13	20%	20%	35%	35%
14	50%	50%	50%	50%
15	35%	35%	35%	35%
16	35%	20%	35%	50%
17	10%	20%	35%	35%
18	10%	35%	35%	35%
19	35%	35%	20%	20%
20	5%	5%	35%	35%
21	5%	5%	35%	35%
22	10%	5%	35%	50%
23	20%	10%	35%	50%